

DEFIANCE, OH, USA

WMO: 725254

Lat: 41.338N

Lon: 84.429W

Elev: 216

StdP: 98.76

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04851

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-14.8	-22.0	0.5	-17.3	-18.9	0.7	-14.0	9.9	-0.7	9.0	-1.1	2.3	240	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	32.6	23.1	31.3	22.6	29.9	21.7	25.0	30.5	24.0	28.9	23.2	27.7	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.3	27.7	22.5	17.6	27.0	21.9	17.0	26.2	76.7	30.4	73.1	28.9	69.8	27.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.9	7.1	-21.6	34.7	3.7	2.1	-24.3	36.2	-26.4	37.5	-28.5	38.6	-31.2	40.2		
(5)			-21.9	26.2	3.6	1.1	-24.5	26.9	-26.6	27.6	-28.6	28.2	-31.3	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-3.6	-2.1	3.3	10.1	16.2	21.2	23.2	22.0	18.5	12.0	5.5	-0.5	(6)
(7)		DBStd	10.65	6.60	6.11	6.04	5.10	4.73	3.64	3.18	2.77	4.13	4.97	5.22	5.90	(7)
(8)		HDD10.0	1568	421	341	223	59	5	0	0	0	1	36	154	328	(8)
(9)		HDD18.3	3309	678	573	467	250	98	14	2	4	47	206	387	583	(9)
(10)		CDD10.0	1771	1	2	16	63	198	336	409	372	256	97	18	3	(10)
(11)		CDD18.3	471	0	0	1	4	34	101	152	118	53	8	0	0	(11)
(12)		CDH23.3	4324	0	0	9	50	335	904	1471	981	496	77	2	0	(12)
(13)		CDH26.7	1428	0	0	2	6	96	310	559	285	156	15	0	0	(13)
(14)	Wind	WSAvg	3.0	3.6	3.5	3.6	3.9	3.1	2.6	2.2	2.1	2.3	2.9	3.4	3.4	(14)
(15)	Precipitation	PrecAvg	902	60	56	61	83	100	92	93	89	75	69	64	62	(15)
(16)		PrecMax	1116	131	140	126	150	163	175	173	260	150	175	134	140	(16)
(17)		PrecMin	679	21	13	19	17	33	26	46	24	12	8	26	13	(17)
(18)		PrecStd	115	29	33	25	32	42	36	38	59	44	46	27	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	17.6	24.0	27.5	31.4	33.9	35.4	32.9	32.7	28.8	21.9	17.2	(19)
(20)			MCWB	12.9	12.1	17.3	18.3	21.2	22.9	24.5	23.2	22.4	21.0	15.3	14.6	(20)
(21)		2%	DB	11.5	12.7	19.1	24.6	29.2	32.1	32.9	31.2	30.9	25.4	18.2	13.6	(21)
(22)			MCWB	9.5	10.5	14.0	16.1	19.8	22.3	23.9	23.0	21.8	17.9	13.6	11.5	(22)
(23)		5%	DB	8.3	9.1	16.1	22.2	27.3	30.1	31.5	29.5	28.1	22.8	16.2	10.5	(23)
(24)			MCWB	6.4	6.7	11.9	14.9	19.2	21.5	23.0	22.1	20.4	17.1	12.3	8.4	(24)
(25)		10%	DB	5.0	6.0	12.9	19.2	24.8	28.1	29.9	28.0	26.3	20.6	13.7	7.4	(25)
(26)			MCWB	3.3	3.8	9.3	13.2	18.0	20.7	22.1	21.4	19.3	15.7	10.5	5.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	13.9	17.6	19.9	22.9	25.0	26.1	25.7	24.4	21.9	16.8	14.8	(27)
(28)			MCD B	15.6	16.0	23.9	25.1	28.8	31.4	32.0	30.8	30.4	27.3	20.3	16.9	(28)
(29)		2%	WB	10.0	10.4	14.9	17.7	21.4	23.8	25.3	24.5	23.0	19.9	14.6	11.8	(29)
(30)			MCD B	11.2	12.8	18.4	22.3	27.1	29.4	31.2	29.0	28.1	23.6	17.4	13.2	(30)
(31)		5%	WB	6.6	6.9	12.6	15.8	20.3	22.8	24.4	23.6	21.9	18.0	12.8	8.5	(31)
(32)			MCD B	8.4	8.9	15.5	20.6	25.4	28.2	29.5	27.5	26.4	21.3	15.9	10.1	(32)
(33)		10%	WB	3.3	3.8	9.7	14.1	19.1	21.9	23.4	22.6	20.7	16.3	10.7	5.7	(33)
(34)			MCD B	5.0	5.8	12.8	18.8	23.6	26.5	27.9	26.4	24.5	20.1	13.5	7.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	8.3	9.7	11.5	11.5	11.1	11.3	11.1	12.1	10.8	8.8	7.1	(35)
(36)			MCDBR	9.9	11.6	13.9	15.3	14.2	13.1	12.4	12.4	14.3	13.5	12.2	9.8	(36)
(37)			MCWBR	8.4	9.6	9.5	8.8	7.1	5.7	4.9	5.6	6.6	7.5	8.7	8.3	(37)
(38)		5% WB	MCD BR	9.4	11.3	12.4	13.1	12.2	10.9	10.8	10.2	11.4	11.2	10.6	9.2	(38)
(39)			MCWBR	8.5	9.9	9.4	8.9	6.9	5.6	5.0	5.3	5.8	7.3	8.6	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.335	0.352	0.401	0.434	0.446	0.439	0.437	0.396	0.355	0.324	0.307	(40)	
(41)		taud	2.382	2.284	2.335	2.214	2.170	2.181	2.219	2.210	2.303	2.405	2.443	2.412	(41)	
(42)		Ebn at Noon	844	865	894	869	845	833	835	825	837	835	816	810	(42)	
(43)		Edh at Noon	82	106	113	137	147	146	139	136	116	93	77	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.57	2.50	3.52	4.64	5.39	6.02	6.07	5.33	4.36	2.90	1.86	1.32	(44)	
(45)		RadStd	0.17	0.34	0.31	0.41	0.49	0.41	0.33	0.24	0.36	0.25	0.26	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	N/A

Nomenclature: See separate page